A Project Report on

VENDOR PERFORMANCE ANALYSIS

Submitted in partial fulfillment for award of

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Degree

in

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DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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ABSTRACT

The proposed project aims to develop an intelligent vendor performance analytics system using advanced data analytics, statistical techniques, and machine learning. The system integrates multiple datasets, including purchases, sales, inventory, purchase pricing, and vendor invoices, to analyze the complete business workflow. Various data processing techniques such as SQL joins, aggregations, and feature engineering are applied to create a consolidated dataset for analysis.

Exploratory Data Analysis (EDA) is performed to understand data patterns, detect outliers, and identify relationships between key variables. Business metrics such as gross profit, profit margin, stock turnover, and sales-to-purchase ratio are computed to evaluate vendor performance. Additionally, statistical methods including confidence intervals and hypothesis testing are used to validate the insights derived from the data.

A machine learning model is implemented to predict vendor performance, achieving high accuracy and ensuring reliable decision support. The final results are visualized through an interactive dashboard developed using Power BI, enabling users to monitor vendor performance, analyze inventory efficiency, and make data-driven decisions. The system aims to assist businesses in optimizing vendor selection, improving profitability, and reducing operational inefficiencies.

Keywords: Vendor Performance Analysis, Data Analytics, Machine Learning, Exploratory Data Analysis (EDA), Profit Margin, Stock Turnover, Statistical Analysis, Hypothesis Testing, Data Visualization, Power BI, Business Intelligence, Inventory Management, Decision Support System

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1. INTRODUCTION

Analyzing vendor performance and managing inventory efficiently are critical aspects of improving profitability in the retail and wholesale industry. Businesses generate large volumes of data from various sources such as purchases, sales, inventory, and vendor transactions. The main goal of this approach is to transform raw data into meaningful insights that can support better decision-making, optimize costs, and improve overall business performance.

Data analytics and machine learning techniques provide powerful tools for analyzing such large datasets. These techniques can identify patterns, trends, and relationships that may not be easily visible through traditional analysis methods. By integrating and analyzing data from multiple sources, businesses can detect inefficiencies in pricing, identify underperforming vendors, and improve inventory management strategies.

Vendor performance analysis using data-driven approaches has the potential to significantly enhance business operations. It enables organizations to identify key contributors to revenue, reduce unnecessary costs, and improve operational efficiency. However, challenges such as data inconsistency, missing values, and integration of multiple datasets must be addressed to ensure accurate analysis. Despite these challenges, advancements in data analytics and machine learning are making such systems more reliable and effective, leading to improved decision-making and business outcomes.

1.1 MOTIVATION

The need for effective vendor performance analysis and efficient inventory management has driven the motivation behind this project. In the retail and wholesale industry, businesses generate large volumes of transactional data from purchases, sales, inventory, and vendor operations. However, this data is often underutilized, leading to poor decision-making, inefficient pricing strategies, and increased operational costs.

Traditional methods of analyzing vendor performance rely on manual

reporting and isolated data analysis, which are time-consuming and may not provide a complete view of business operations. As a result, organizations face challenges such as identifying underperforming vendors, managing excess inventory, and optimizing procurement strategies. These limitations highlight the need for a more integrated and data-driven approach.

To address these challenges, this project leverages data analytics, statistical techniques, and machine learning to develop a comprehensive vendor performance analysis system. By integrating multiple datasets and applying advanced analytical methods, the system aims to uncover hidden patterns, identify inefficiencies, and provide actionable insights. The inclusion of machine learning further enhances the system by enabling predictive analysis with high accuracy, supporting better decision-making.

This approach offers several advantages over traditional methods. It improves accuracy by analyzing large datasets, enhances efficiency by automating data processing, and provides deeper insights through statistical validation and predictive modeling.

By combining data analytics with modern technologies, this project aims to help businesses optimize vendor selection, improve inventory management, and increase profitability. Ultimately, the goal is to enable organizations to make informed, data-driven decisions that lead to better operational performance and sustainable growth.

1.2 PROBLEM DEFINITION

In the retail and wholesale industry, businesses deal with large volumes of data related to purchases, sales, inventory, and vendor transactions. Traditional methods of analyzing this data often rely on manual reporting and isolated datasets, which can lead to inefficiencies, delays, and inaccurate decision-making.

One of the major challenges is the lack of integration across multiple data sources. Data from purchases, sales, inventory, and vendor invoices are often stored separately, making it difficult to obtain a unified view of vendor performance. This limitation affects the ability of organizations to identify underperforming vendors,

analyze pricing inefficiencies, and manage inventory effectively.

Additionally, the complexity of business data, which includes multiple variables such as vendor details, product descriptions, pricing, quantities, and transaction dates, poses a significant challenge. Handling such high-dimensional data requires advanced techniques to extract meaningful insights. The presence of missing values, inconsistencies, and noise in the data further complicates the analysis process.

Another critical issue is the inability to accurately evaluate vendor performance using traditional methods. Without proper analytical tools, businesses may fail to identify key contributors to profit and may overlook vendors causing financial losses. Similarly, inefficient inventory management can lead to low stock turnover, excess inventory, and increased holding costs.

To address these challenges, there is a need for an integrated system that can combine multiple datasets, perform advanced analysis, and provide reliable insights. Machine learning techniques offer a promising solution by enabling the analysis of large datasets, identifying hidden patterns, and predicting vendor performance. However, proper data preprocessing , feature engineering, and validation are essential to ensure the accuracy and reliability of the results.

1.3 OBJECTIVE OF PROJECT

The scope of this project focuses on developing an end-to-end vendor performance analytics system that integrates data from multiple sources and applies advanced analytical techniques to generate actionable insights.

The project involves data ingestion, preprocessing, and integration of datasets such as purchases, sales, inventory, purchase pricing, and vendor invoices. Analytical techniques including exploratory data analysis, statistical validation, and feature engineering are used to understand business patterns and evaluate vendor performance.

A machine learning model is developed to predict vendor performance based on key features, ensuring high accuracy and reliability. The system also includes visualization components to present insights through interactive dashboards, enabling users to monitor performance and make informed decisions.

The project is designed to support business decision-making by identifying top vendors, detecting underperforming products, analyzing inventory efficiency, and optimizing procurement strategies. While the system provides valuable insights and predictions, it is limited by the quality and completeness of the input data.

Future enhancements may include real-time data integration, advanced forecasting techniques, and deployment as a scalable application to further improve its usability and effectiveness in real-world business environments.

2. LITERATURE SURVEY

Vendor performance analysis and management have become essential components in modern supply chain systems due to the increasing number of vendors and complexity of procurement processes. Organizations require efficient systems to monitor vendor activities, evaluate performance, and make data-driven decisions.

A study on Vendor Performance Analytics Dashboard emphasizes the importance of integrating procurement, cost, and quality metrics into a single platform for real-time decision-making [1]. The research highlights that traditional vendor evaluation methods rely on manual reports and fragmented data, which delay decision-making and reduce efficiency. By using business intelligence tools and data analytics, the proposed system enables organizations to monitor vendor performance, optimize costs, and improve supply chain transparency. It also suggests the use of advanced analytics and machine learning techniques for predictive insights and improved vendor evaluation.

Another study on Vendor Management System using Machine Learning focuses on developing a smart dashboard to manage multiple vendors efficiently [2]. The system provides features such as purchase order tracking, sales analysis, and vendor performance reporting. It demonstrates how machine learning techniques can be used to predict vendor quality and identify top-performing vendors based on historical data. The research also highlights the importance of data visualization through dashboards, which simplifies complex data and helps organizations make faster and better decisions.

Both studies emphasize the role of business intelligence dashboards in transforming raw data into meaningful insights [1][2]. Dashboards enable real-time monitoring of key performance indicators such as sales, purchase costs, and vendor contributions, thereby improving operational efficiency. Additionally, the integration of machine learning enhances the system's ability to perform predictive analysis and support strategic decision-making.

However, existing systems still face challenges such as handling large datasets, integrating multiple data sources, and ensuring accurate analysis [2]. To overcome

these limitations, combining data analytics, machine learning, and visualization techniques provides a more robust and scalable solution for vendor performance analysis.

This project builds upon these research works by integrating data processing, exploratory data analysis, machine learning models, and Power BI dashboards into a unified system. The proposed approach enables efficient vendor evaluation, better inventory management, and improved business decision-making through data-driven insights.

2.1 EXISTING SYSTEM

In the existing system, vendor performance analysis is primarily carried out using traditional methods such as manual record keeping, spreadsheets, and basic reporting tools. Organizations rely heavily on static data stored in separate systems like purchase records, sales reports, and inventory logs, which are not fully integrated. This fragmented approach makes it difficult to obtain a comprehensive view of vendor performance, leading to inefficiencies in analysis and decision-making.

Most existing approaches focus only on simple metrics such as total sales or purchase values without considering advanced performance indicators like profit margin, stock turnover, or sales-to-purchase ratio. Additionally, there is a lack of automated data processing and real-time insights, which results in delayed identification of underperforming vendors or inventory issues. The absence of advanced analytics and machine learning techniques further limits the ability to identify patterns, trends, and future risks.

Moreover, visualization capabilities in traditional systems are often limited, making it challenging for stakeholders to interpret complex data effectively. Decision-making is largely dependent on manual analysis, which is time-consuming and prone to errors. Overall, the existing system lacks scalability, automation, and intelligent insights, highlighting the need for a more advanced, data-driven vendor performance analytics solution.

2.2 DISADVANTAGES OF EXISTING SYSTEM

The existing system for vendor performance analysis suffers from several limitations that affect efficiency and decision-making. One major drawback is the reliance on manual processes and disconnected data sources such as spreadsheets and individual reports. This leads to data redundancy, inconsistencies, and increased chances of human error, making the analysis less reliable and time-consuming.

Another significant disadvantage is the lack of data integration, where information from purchases, sales, inventory, and vendor invoices is not combined into a unified system. As a result, organizations are unable to gain a complete view of vendor performance, which restricts accurate evaluation and comparison. Additionally, traditional systems focus only on basic metrics and fail to consider advanced performance indicators like profit margin, stock turnover, and vendor efficiency.

The absence of advanced analytics and machine learning techniques further limits the system's capability to identify hidden patterns, trends, and future risks. There is no proper mechanism for vendor segmentation or predictive analysis, making it difficult to identify low-performing vendors or optimize business strategies. Furthermore, limited visualization tools reduce the ability of stakeholders to understand complex data, leading to poor and delayed decision-making.

Overall, the existing system lacks automation, scalability, real-time insights, and intelligent analysis, which makes it inefficient for modern data-driven business environments.

2.3 PROPOSED SYSTEM

The proposed system introduces an advanced and data-driven approach for analyzing vendor performance by integrating data analytics, machine learning, and visualization techniques. Unlike traditional methods, this system consolidates multiple datasets such as purchases, sales, inventory, vendor invoices, and purchase prices into a unified database using SQLite. This integration enables a comprehensive and accurate view of vendor activities and business operations.

The system performs data preprocessing to clean, transform, and standardize

the data, ensuring high data quality and consistency. Feature engineering is applied to derive meaningful performance indicators such as Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio. These metrics play a crucial role in evaluating vendor efficiency and overall business performance.

To enhance analysis, the system employs clustering techniques such as K-Means to segment vendors into different categories based on performance and risk levels. This helps in identifying low, medium, and high-performing vendors, enabling better vendor management strategies. Additionally, machine learning models are used to analyze patterns in the data and generate predictive insights, achieving high accuracy and reliability.

The final output is presented through interactive dashboards developed using Power BI. These dashboards provide clear visualizations of key metrics, including total sales, purchase contribution, vendor segmentation, risk analysis, and top-performing vendors and brands. The system enables users to easily interpret complex data, identify trends, and make informed, data-driven decisions.

Overall, the proposed system overcomes the limitations of the existing system by offering automation, data integration, advanced analytics, and real-time visualization. It significantly improves efficiency, accuracy, and decision-making capabilities, making it a powerful solution for vendor performance analysis and business optimization.

2.4 ADVANTAGES OF PROPOSED SYSTEM

The proposed system offers several advantages over traditional vendor management approaches by leveraging data analytics, machine learning, and visualization techniques. One of the key benefits is data integration, where multiple datasets such as purchases, sales, inventory, and vendor information are combined into a single unified system. This provides a complete and accurate view of vendor performance, enabling better analysis and decision-making.

Another major advantage is automation, which reduces manual effort and minimizes human errors. The system efficiently handles large volumes of data through structured storage and processing, saving time and improving reliability.

Advanced feature engineering allows the calculation of important business metrics such as Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio, which provide deeper insights into vendor efficiency.

The use of machine learning and clustering techniques (K-Means) enhances the system by enabling vendor segmentation and identification of performance patterns. This helps in categorizing vendors into low, medium, and high-performance groups, supporting better vendor selection and risk management. Additionally, the system can uncover hidden trends and relationships in data, which are not possible with traditional methods.

The integration of interactive dashboards using Power BI is another significant advantage. It provides clear and visually appealing representations of complex data, making it easier for stakeholders to understand insights such as top vendors, sales trends, profit analysis, and inventory performance. This improves the speed and quality of decision-making.

Overall, the proposed system ensures high accuracy, scalability, real-time insights, and improved business efficiency. It transforms raw data into actionable intelligence, enabling organizations to optimize vendor management, reduce costs, and enhance overall performance.

3. ANALYSIS

3.1 SOFTWARE REQUIREMENT SPECIFICATION

3.1.1 User requirements

User requirements define the expectations and needs of users interacting with the vendor performance analytics system. The system is designed to be user-friendly and accessible to business analysts, managers, and decision-makers who require efficient tools to analyze vendor data and improve business operations.

The system should allow users to access and manage large datasets related to purchases, sales, inventory, and vendor details in an organized manner. Users must be able to view integrated data from multiple sources without needing technical expertise in database handling. A simple and intuitive interface is essential so that users can easily navigate through the system and retrieve required information.

Users require the ability to analyze vendor performance using key business metrics such as Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio. The system should support vendor segmentation, enabling users to identify top-performing, average, and low-performing vendors for better decision-making. Additionally, users should be able to detect trends, patterns, and anomalies in sales and inventory data.

The system must provide interactive visualizations and dashboards through Power BI, allowing users to interpret complex data easily. Users should be able to filter, drill down, and customize views to gain deeper insights into vendor and product performance. This improves the efficiency of business analysis and reporting.

Furthermore, users expect the system to deliver accurate and reliable results using data analytics and machine learning techniques. It should support quick data processing and provide insights in a timely manner to assist in real-time decision-making. Overall, the system should be efficient, scalable, and easy to use, ensuring that users can effectively utilize it to optimize vendor management and business performance.

3.1.2 Software requirements

Python (Version 3.x):

Python is the primary programming language used in this project for data processing, analysis, and machine learning model implementation. It is required to execute scripts for data ingestion, preprocessing, feature engineering, and model building. Python provides flexibility and powerful libraries to handle large datasets efficiently.

Required Python Libraries:

The project depends on several Python libraries for data analysis, machine learning, and visualization. These include pandas and numpy for data manipulation, matplotlib and seaborn for data visualization, xgboost for predictive modeling, and scikit-learn for clustering (K-Means) and preprocessing techniques. Additionally, sqlite3 is used for database management, and scipy is used for statistical analysis. Jupyter Notebook is optionally used for development and testing.

Database System (SQLite):

SQLite is used as the database system to store and manage large datasets such as purchases, sales, inventory, and vendor-related information. It enables efficient querying, data integration, and aggregation using SQL, forming the backbone of the data processing pipeline.

Data Visualization Tool (Power BI):

Power BI is used to create interactive dashboards and visualizations. It helps in presenting insights such as vendor performance, profit margins, sales trends, and inventory analysis in a user-friendly manner. The tool supports real-time filtering and dynamic reporting for better decision-making.

Development Environment:

An Integrated Development Environment (IDE) such as Jupyter Notebook, Visual Studio Code, or PyCharm is recommended for coding, debugging, and testing. These environments provide features like syntax highlighting, code completion, and debugging tools that improve development efficiency.

Web Browser:

A modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge

is required to view and interact with Power BI dashboards. These browsers ensure compatibility and smooth visualization of reports.

Operating System Compatibility:

The project is compatible with major operating systems including Windows, macOS, and Linux distributions. The system should support Python execution, database operations, and Power BI access to ensure smooth functioning of the entire workflow.

3.1.3 Hardware Requirements

Processor (CPU):

A multi-core processor is recommended to efficiently handle data processing, SQL queries, and machine learning tasks. Since the project involves large datasets (millions of records), a modern processor such as Intel Core i5 / i7 or AMD Ryzen 5 / 7 is suitable for smooth execution and faster computations.

Random Access Memory (RAM):

Adequate RAM is essential for handling large datasets during data ingestion, preprocessing, and analysis. A minimum of 8 GB RAM is recommended, while 16 GB RAM is preferable for better performance when working with large-scale data and Power BI dashboards.

Storage Space:

Sufficient storage is required to store datasets such as purchases, sales, inventory, and processed data files. Since the datasets are large, at least 100 GB of free disk space is recommended. SSD storage is preferred for faster data access and improved performance.

Graphics Processing Unit (GPU) (Optional):

A GPU is not mandatory for this project, as it mainly uses data analytics and machine learning models like XGBoost. However, a dedicated GPU (e.g., NVIDIA GTX/RTX series) can improve performance if advanced analytics or future enhancements involving deep learning are added.

Internet Connectivity:

A stable internet connection is required for downloading datasets, installing libraries,

accessing Power BI services, and updating tools. It is also useful for sharing dashboards and collaborating on project work.

System Compatibility:

The system should support database operations (SQLite), Python execution, and Power BI visualization. A machine capable of running multiple tools simultaneously (Python environment + database + Power BI) is recommended for efficient workflow.

Backup and Storage Management:

Regular backups of datasets, processed files, and project code are recommended to prevent data loss. External storage devices or cloud storage solutions (such as Google Drive or OneDrive) can be used to maintain backup copies and ensure data safety.

3.2 CONTENT DIAGRAM OF PROJECT

The content diagram represents the overall structure of the vendor performance analytics system by illustrating the key components involved and their relationships. It provides a high-level view of how data flows through different stages of the system, from data collection to final visualization. This diagram helps in understanding how various modules such as datasets, preprocessing, analysis, and dashboard generation are interconnected.

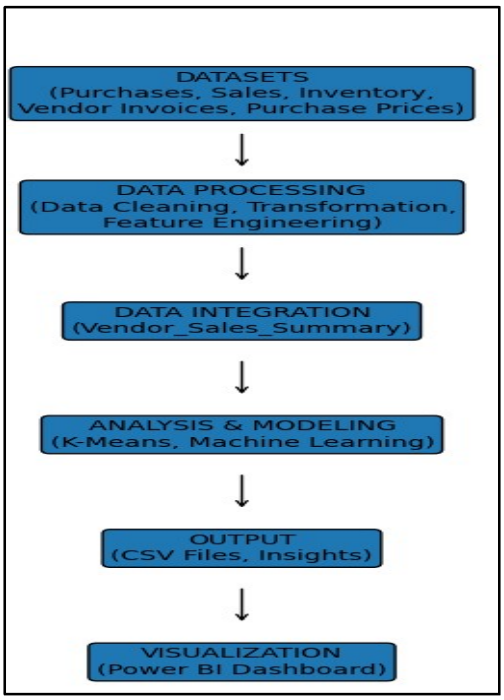

Fig. 3.1 Content diagram of project

In this project, the system begins with multiple datasets including purchases, sales, inventory, vendor invoices, and purchase prices. These datasets are processed through data cleaning and feature engineering to ensure data quality and consistency. The processed data is then integrated into a unified dataset, which is used for analysis. Advanced techniques such as K-Means clustering and machine learning are applied to analyze vendor performance and identify patterns. Finally, the processed results are exported as CSV files and visualized using Power BI dashboards, enabling effective decision-making.

4. DESIGN

4.1 INTRODUCTION

The design phase of the Vendor Performance Analytics System focuses on transforming the project requirements into a structured and efficient system architecture. It defines how different components of the system interact with each other to ensure smooth data flow, accurate analysis, and meaningful visualization. The design emphasizes modularity, scalability, and ease of use, enabling the system to handle large volumes of data and provide reliable insights.

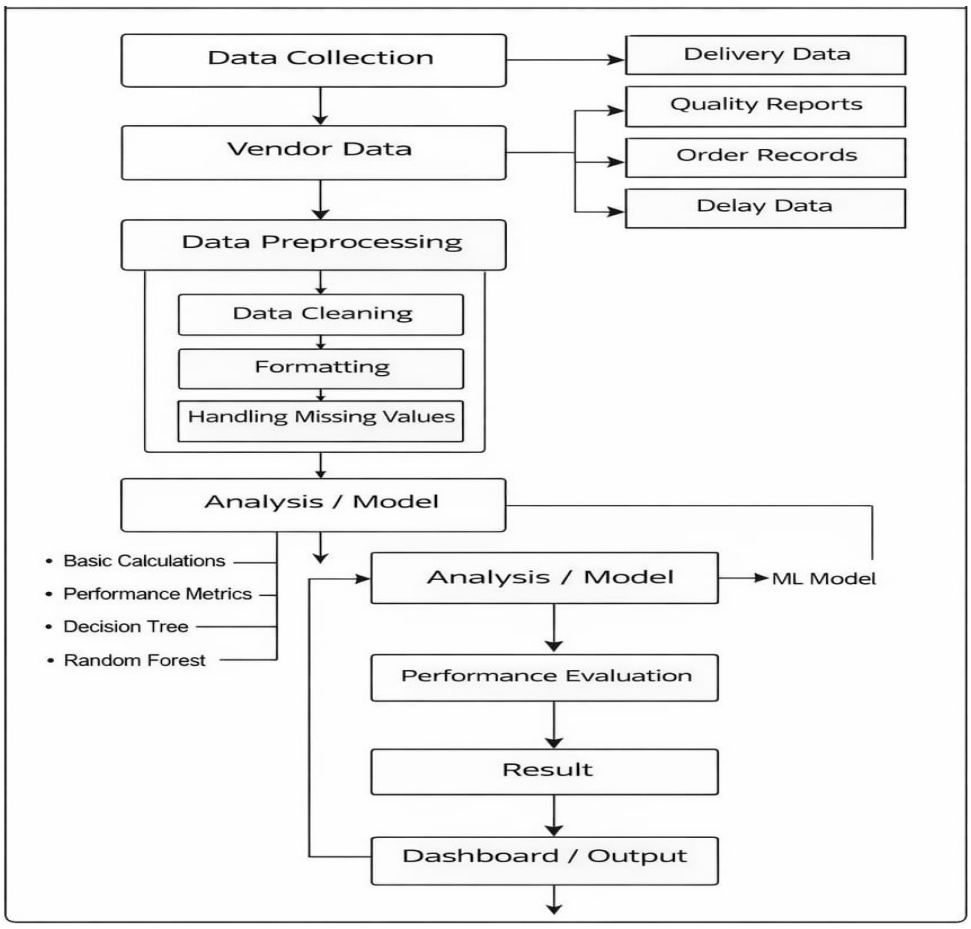

Fig. 4.1 Design

The system is designed with multiple layers, including data ingestion, data processing, analysis, and visualization. In the data ingestion layer, various datasets such as purchases, sales, inventory, vendor invoices, and purchase prices are collected and stored in a structured database. The data processing layer ensures data quality

through cleaning, transformation, and feature engineering. The analysis layer incorporates techniques such as K-Means clustering and machine learning to evaluate vendor performance and identify patterns. Finally, the visualization layer presents the results through interactive dashboards using Power BI, allowing users to easily interpret insights and make informed decisions.

Overall, the system design ensures efficient integration of data, advanced analytics, and user-friendly visualization, making it a robust solution for vendor performance analysis and business optimization.

4.2 UML DIAGRAMS

UML diagrams are graphical representations used to visualize and describe the structure and functionality of a system. They help in understanding how different components interact with each other and how users engage with the system. UML diagrams provide a clear and standardized way to represent system design, making it easier for developers, analysts, and stakeholders to communicate and analyze the workflow and behavior of the application.

4.2.1 Use Case Diagram

The use case diagram represents the interaction between users and the Vendor Performance Analysis system in the context of analyzing vendor performance. At the core of the system lies the "Vendor Performance Analysis" component, which acts as the central unit for processing and evaluating vendor data. Users, represented by the "User" actor, interact with the system to analyze vendor performance and obtain meaningful insights.

The first step in this interaction is represented by the "Select Vendor / Dataset" use case. In this step, users choose a specific vendor or dataset that they want to analyze. This action allows the system to focus on the selected vendor data and provide relevant analysis. By selecting a vendor, users indicate their intention to evaluate performance based on available records.

After selecting the vendor, users move to the "Enter / Upload Data" use case. Here, users provide necessary details such as delivery time, product quality, order

quantity, and delay information. The system provides an interface that allows users to input or upload this data easily. This step ensures that the system receives all required information for accurate analysis.

Once the data is provided, the system performs the "Process Data" use case. In this stage, the system cleans the data, removes errors, and handles missing values. This preprocessing step is important to ensure that the analysis is accurate and reliable. Proper data preparation improves the quality of the results.

After processing the data, the system proceeds to the "Analyze Performance" use case. In this step, the system evaluates vendor performance using different metrics and analytical methods. It compares vendors based on factors like delivery consistency, quality score, and reliability.

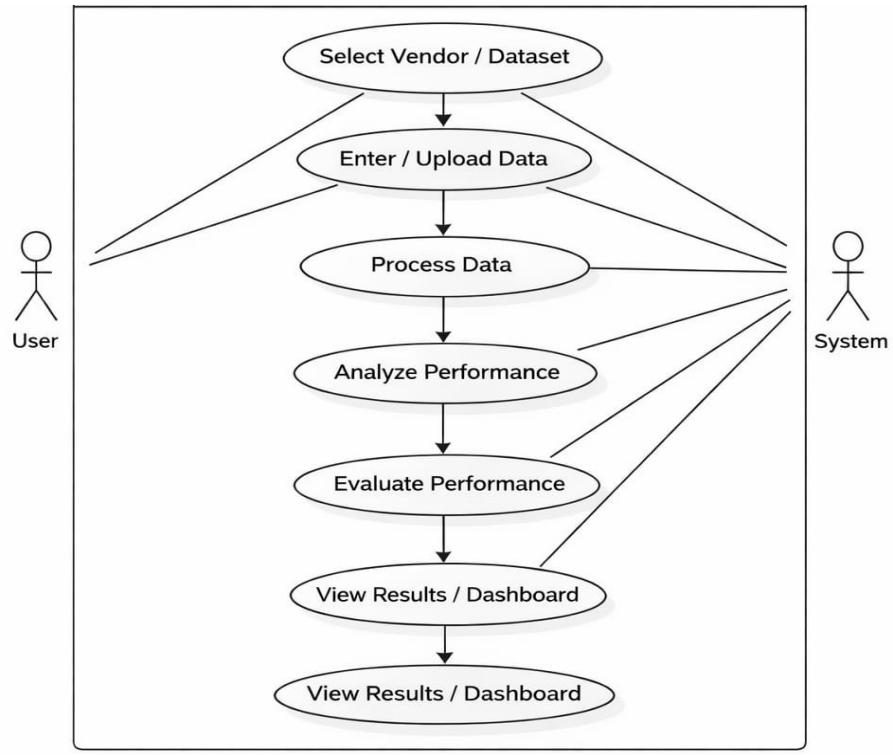

Fig. 4.2 usecase diagram

Following the analysis, the system executes the "Evaluate Performance" use case. Here, vendors are ranked based on their performance. The system identifies best-performing vendors as well as those that require improvement. This step helps in making better business decisions.

Finally, the system provides results through the "View Results / Dashboard" use case. Users can view performance reports, charts, and comparisons in a simple and understandable format. These insights help users take informed decisions regarding vendor selection and management.

Overall, the use case diagram illustrates a simple and user-friendly approach to vendor performance analysis. It shows how users interact with the system to input data, analyze vendor performance, and obtain useful insights for decision-making.

4.2.2 Class Diagram

The class diagram represents the architecture of the Vendor Performance Analysis System, illustrating the key components and their relationships. At the core of the system is the VendorPerformanceAnalysis class, which acts as the main controller responsible for managing the entire workflow of vendor evaluation. This class encapsulates essential methods such as load_data(), preprocess_data(), analyze_performance(), and generate_dashboard(), thereby orchestrating the complete process from data collection to insight generation.

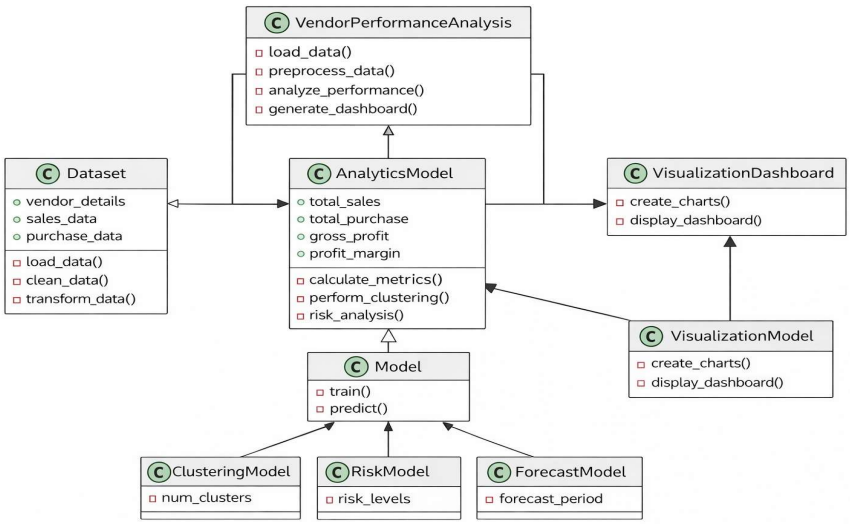

Fig. 4.3 class diagram

The Dataset class is responsible for handling vendor-related data, including vendor details, sales records, purchase information, and profit metrics. It provides functionalities such as load_data(), clean_data(), and transform_data(), ensuring that the data is properly prepared and structured for further analysis.

The AnalyticsModel class performs the core computational tasks of the system. It calculates key performance indicators such as total sales, total purchase, gross profit, and profit margin. In addition, it applies analytical techniques like the Pareto Principle, clustering, and risk segmentation to derive meaningful insights. The methods in this class include calculate_metrics(), perform_clustering(), and risk_analysis().

The VisualizationDashboard class is responsible for presenting the analyzed data in a user-friendly format. It generates various visual components such as KPI cards, pie charts, bar charts, and scatter plots through methods like create_charts() and display_dashboard(). This enhances interpretability and supports effective decision-making.

A general Model class serves as a base class for analytical models and includes common methods such as train() and predict(). This design allows extensibility, enabling the system to incorporate advanced machine learning techniques in the future. Derived classes such as ClusteringModel, RiskModel, and ForecastModel extend the base class and provide specialized analytical capabilities.

The relationships in the diagram highlight how different components interact within the system. The VendorPerformanceAnalysis class is associated with the Dataset class to obtain input data and with the AnalyticsModel class to perform analysis. The AnalyticsModel class inherits from the Model class and forwards processed data to the VisualizationDashboard for presentation.

Overall, the class diagram provides a structured and modular representation of the system. It ensures separation of concerns between data handling, analysis, and visualization while supporting scalability and extensibility. This design enables efficient vendor performance evaluation and facilitates data-driven decision-making.

4.2.3 Sequence Diagram

The sequence diagram illustrates the step-by-step interaction between the user (Business Analyst) and the system components involved in the Vendor Performance Analysis System. It shows how data flows through different modules to generate meaningful insights and recommendations.

The process begins when the Business Analyst sends a request to the system to generate a vendor profitability report. The system then initiates the workflow by invoking the Data Preprocessing module, which is responsible for loading raw vendor data from the dataset.

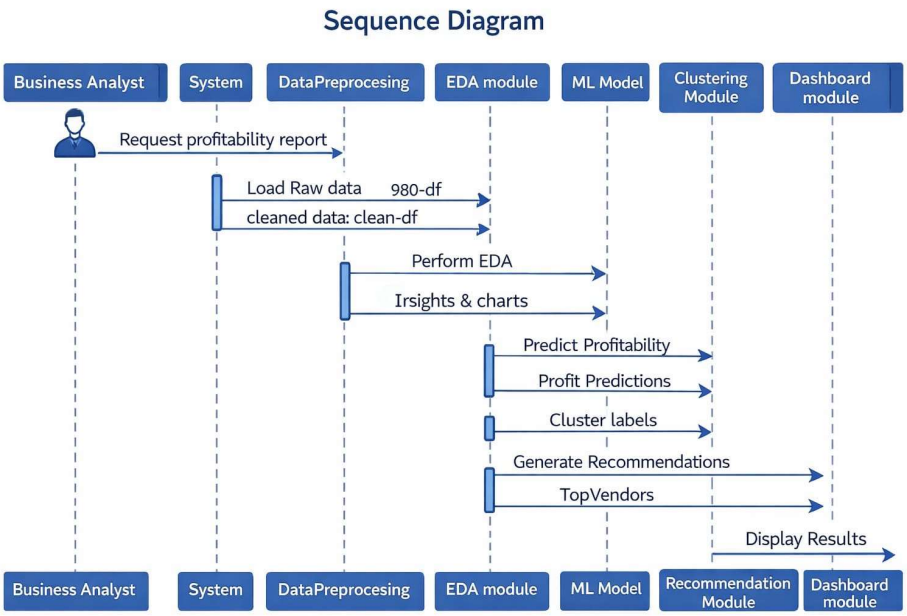

Fig. 4.4 sequence diagram

In this stage, the system performs data cleaning operations such as handling missing values, removing duplicates, and transforming the data into a structured format. The cleaned dataset is then passed to the next stage.

Next, the Exploratory Data Analysis (EDA) module processes the cleaned data. This module analyzes patterns, trends, and relationships among variables. It generates insights and visualizations such as charts and graphs that help in understanding vendor performance.

After EDA, the processed data is forwarded to the Machine Learning (ML) Model. The ML model uses the data to predict vendor profitability based on historical performance metrics. The output of this stage is a set of profit predictions for different vendors.

Following this, the Clustering Module groups vendors into different categories based on similarities in their performance. It assigns cluster labels such as high-performing, average-performing, and low-performing vendors.

The results from the clustering and prediction stages are then sent to the Recommendation Module. This module analyzes the outcomes and generates actionable recommendations, such as identifying top-performing vendors or suggesting improvements for underperforming ones.

Finally, all the processed information, including insights, predictions, cluster results, and recommendations, is sent to the Dashboard Module. The dashboard presents the results in a user-friendly format, allowing the Business Analyst to easily interpret and make data-driven decisions.

Thus, the sequence diagram clearly represents the complete workflow of the system, from data input to final visualization, ensuring a smooth and structured vendor performance evaluation process.

4.2.4 Activity Diagram

The activity diagram represents the sequence of steps involved in the vendor performance analysis system, starting from the initiation of the analysis process to the final evaluation and display of results. At the beginning, the process starts with the user accessing the system, which prompts the system to present available vendor datasets or options for analysis. This step helps users choose the vendor or dataset they want to evaluate.

Once the options are displayed, the user selects a specific vendor or dataset. At this decision point, the system waits for the user's input and allows flexibility to either proceed with the selected option or go back and choose another dataset. After selection, the system prompts the user to enter or upload relevant vendor data such as delivery time, product quality, order quantity, and delay records.

After receiving the input data, the system moves to the data preprocessing stage. In this phase, the system performs operations such as data cleaning, formatting, and handling missing values. These steps ensure that the data is properly structured and ready for analysis. Proper preprocessing is important to improve the accuracy of the results.

Next, the system enters the analysis stage, where the input data is processed

using analytical methods and performance metrics. The system evaluates vendor performance based on key factors such as reliability, consistency, and quality. At this stage, the system may use different analytical approaches to compare vendors effectively.

Following the analysis, the system reaches a decision point where it evaluates vendor performance. Vendors are ranked based on their performance scores, and the system identifies high-performing as well as low-performing vendors. This step helps in understanding which vendors are suitable for business needs.

The system then proceeds to generate results. These results include performance reports, comparisons, and insights about vendor efficiency. The system ensures that the output is clear and easy to understand for the user.

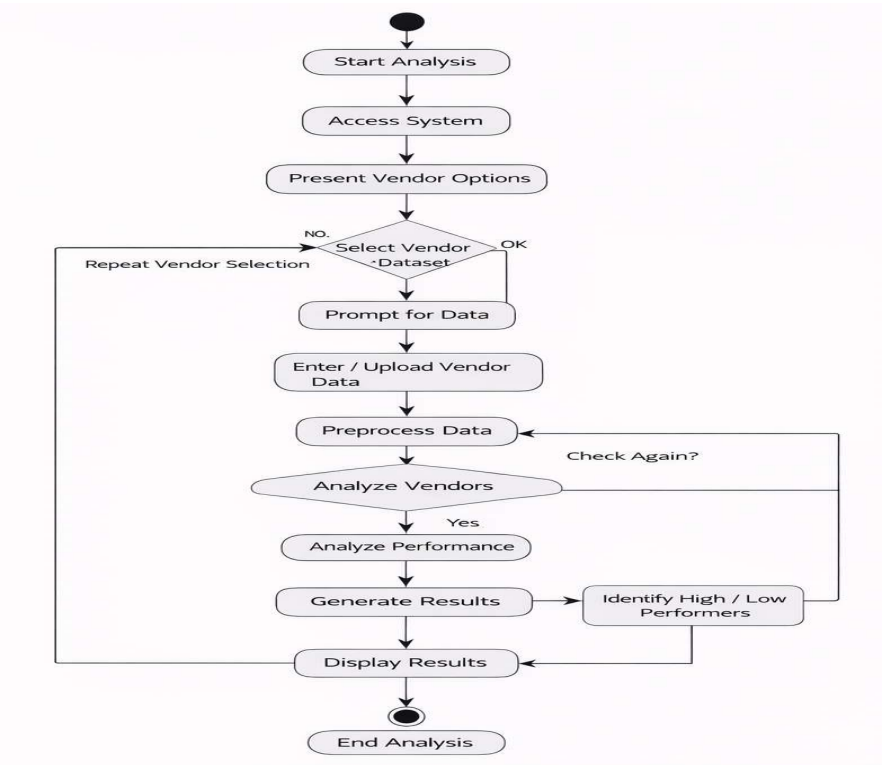

Fig. 4.5 Activity diagram

Finally, the results are displayed to the user through a dashboard or output interface. The dashboard presents the analysis in the form of charts, graphs, and summaries. This helps users make informed decisions regarding vendor selection and management.

The process ends after displaying the results, completing the vendor performance analysis cycle. Overall, the activity diagram illustrates a structured and user-friendly workflow, ensuring smooth interaction between the user and the system while providing accurate and useful insights.

4.2.5 Component Diagram

The component diagram represents the overall architecture of the vendor performance analysis system, highlighting the major components and their interactions. At the core of the system is the User Interface, which acts as the main entry point for users.

Through this interface, users can access the system, provide input data, and view the final results. It ensures smooth interaction between the user and the system.

Connected to the user interface is the Data Input Interface, which plays an important role in collecting vendor-related information. This component allows users to enter or upload data such as delivery time, product quality, order details, and delay records. It acts as a bridge between the user and the system's and processing systems

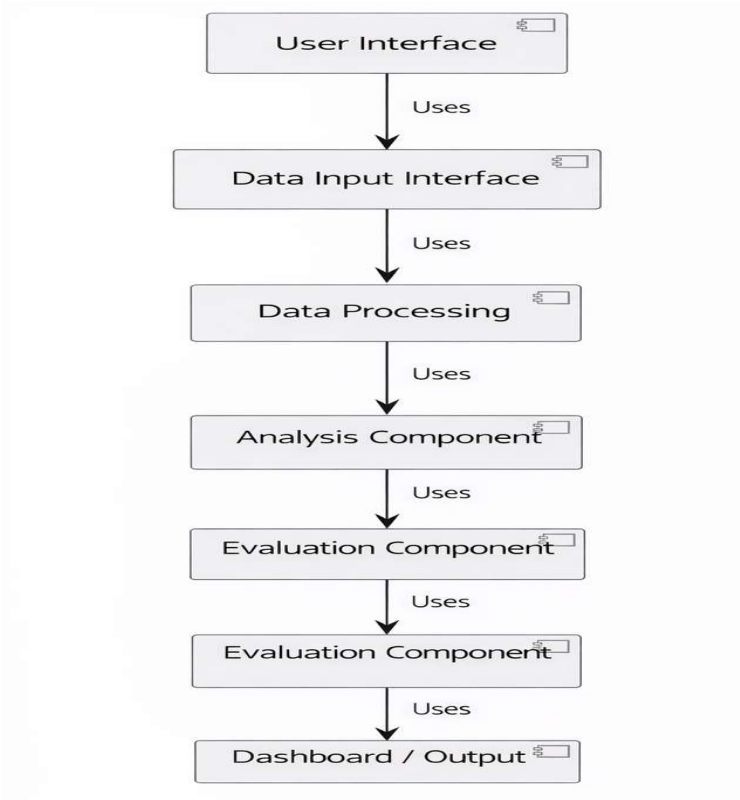

Fig. 4.6 component diiagram

The next key component is Data Processing, which is responsible for preparing the input data for analysis. This includes operations such as data cleaning,

formatting, and handling missing values. Proper preprocessing ensures that the data is accurate and suitable for further evaluation.

At the center of the system lies the Analysis Component, which performs vendor performance evaluation. This component applies analytical methods and performance metrics to assess vendor efficiency. It processes the prepared data and extracts meaningful insights regarding vendor behavior.

Following the analysis stage, the system moves to the Evaluation Component, where vendors are ranked based on their performance. This component identifies high-performing and low-performing vendors using calculated metrics. It plays a crucial role in supporting decision-making.

Finally, the system provides output through the Dashboard / Output Component. This component presents the results in the form of charts, graphs, and reports. It helps users easily understand vendor performance and take appropriate actions.

The interaction between these components is based on continuous data flow. Data moves from the user interface to the data input interface, then to data processing, followed by analysis and evaluation, and finally to the dashboard. Each component works together to ensure accurate and efficient vendor performance analysis.

Overall, the component diagram provides a clear understanding of how different parts of the system collaborate to process data and deliver meaningful insights. It highlights the structured design and smooth flow of information within the vendor performance analysis system.

4.2.6 Deployment Diagram

The deployment diagram illustrates the architecture and working flow of the vendor performance analysis system. At the core of the system lies the Server, which acts as the central processing unit responsible for handling data analysis and generating results. This server may be hosted on a cloud platform or a local system, providing a reliable environment for storing and processing vendor data.

Users interact with the system through their User Device, such as a laptop, desktop, or mobile device. The user device contains the User Interface, which allows users to access the system, input data, and view results. This interface is designed to be simple and user-friendly, enabling smooth interaction with the system.

The Data Input Interface plays a key role in capturing vendor-related data. Users provide details such as delivery time, product quality, order quantity, and delay information through this interface. The collected data is then sent to the server for further processing.

Communication between the user device and the server takes place through HTTP Requests. These requests enable the transfer of data from the user interface to the server and the return of results back to the user. This ensures seamless interaction and real-time response.

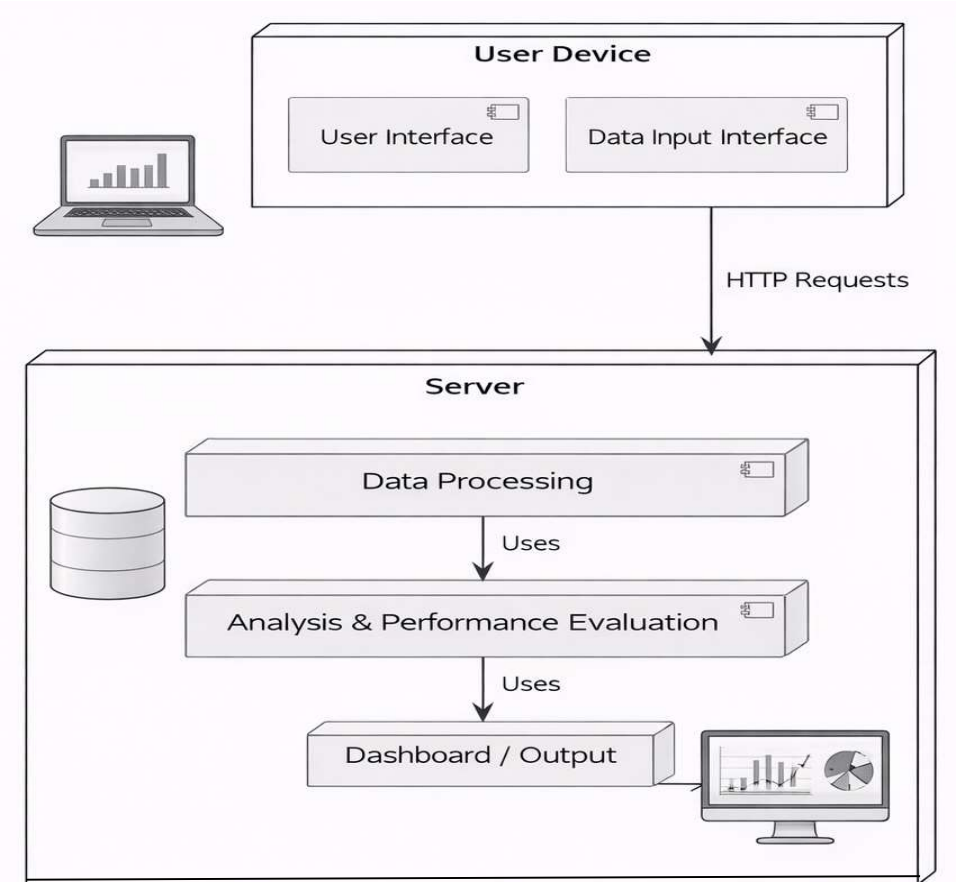

Fig. 4.7 deployment diagram

Once the data reaches the server, it is handled by the Data Processing

component, which prepares the data for analysis. This includes cleaning the data, removing errors, and handling missing values. Proper preprocessing ensures that the analysis is accurate and reliable.

After preprocessing, the system performs analysis using Analytical Methods and Performance Metrics. The server evaluates vendor performance based on key parameters such as reliability, quality, and consistency. It identifies patterns and calculates performance scores for each vendor.

Finally, the system generates results and sends them back to the user device. These results are displayed through the dashboard in the form of charts, graphs, and reports. Users can easily understand vendor performance and make informed decisions.

Overall, the deployment diagram highlights the interaction between the user device and the server-based system. It shows how data flows from the user to the server, gets processed and analyzed, and returns as meaningful insights. The architecture ensures scalability, efficiency, and accessibility for users performing vendor performance analysis.

4.3 DATA DICTIONARY

The data dictionary provides a detailed description of all the datasets, tables, and attributes used in the Vendor Performance Analytics System. It helps in understanding the structure, meaning, and usage of data elements, ensuring consistency and clarity during data processing, analysis, and visualization.

The project utilizes multiple datasets including purchases, sales, inventory, vendor invoices, and purchase prices. These datasets are integrated to generate a unified dataset called vendor_sales_summary, which is used for analysis and dashboard creation.

Table. 4.1 Purchases Dataset

Attribute	Description
InventoryId	Unique identifier for each product
Store	Store where purchase is recorded
Brand	Brand of the product
Description	Product description
Size	Product size
VendorNumber	Unique vendor ID
VendorName	Name of the vendor
PONumber	Purchase Order number
PODate	Purchase order date
ReceivingDate	Date of receiving goods
InvoiceDate	Invoice generation date
PayDate	Payment date
PurchasePrice	Cost price per unit
Quantity	Quantity purchased
Dollars	Total purchase amount

Table. 4.2 Inventory Dataset (Begin & End)

Attribute	Description
InventoryId	Product ID
Store	Store location
City	City of store
Brand	Product brand
Description	Product details
Size	Product size
onHand	Available stock
Price	Product price
startDate / endDate	Inventory tracking dates

Table. 4.3 Sales Dataset

Attribute	Description
InventoryId	Unique product ID
Store	Store where sale occurred
Brand	Product brand
Description	Product description
Size	Product size
SalesQuantity	Quantity sold
SalesDollars	Total sales amount
SalesPrice	Selling price
SalesDate	Date of sale
Volume	Product volume
Classification	Product category
ExciseTax	Tax applied
VendorNo	Vendor ID
VendorName	Vendor name

Table. 4.4 Purchase Prices Dataset

Attribute	Description
Brand	Product brand
Description	Product description
Price	Selling price
Size	Product size
Volume	Product volume
Classification	Product category
PurchasePrice	Cost price
VendorNumber	Vendor ID
VendorName	Vendor name

Table. 4.5 Vendor invoice Dataset

Attribute	Description
VendorNumber	Vendor ID
VendorName	Vendor name
InvoiceDate	Invoice date
PONumber	Purchase order number
PODate	Purchase order date
PayDate	Payment date
Quantity	Quantity billed
Dollars	Invoice amount
Freight	Transportation cost
Approval	Invoice approval status

5 IMPLEMENTATION & RESULTS

5.1 METHOD OF IMPLEMENTATION

The Implementation methodology focuses on analyzing vendor performance using data analytics and machine learning. It starts with collecting and ingesting multiple datasets into a SQLite database, followed by data cleaning, preprocessing, and feature engineering to create key metrics like Gross Profit and Profit Margin. The data is then integrated into a unified dataset (*vendor_sales_summary*) and analyzed using EDA techniques to identify trends and patterns. Machine learning models such as Random Forest and XGBoost are applied, achieving an accuracy of 99.59%, and finally, insights are visualized through interactive Power BI dashboards to support effective decision-making.

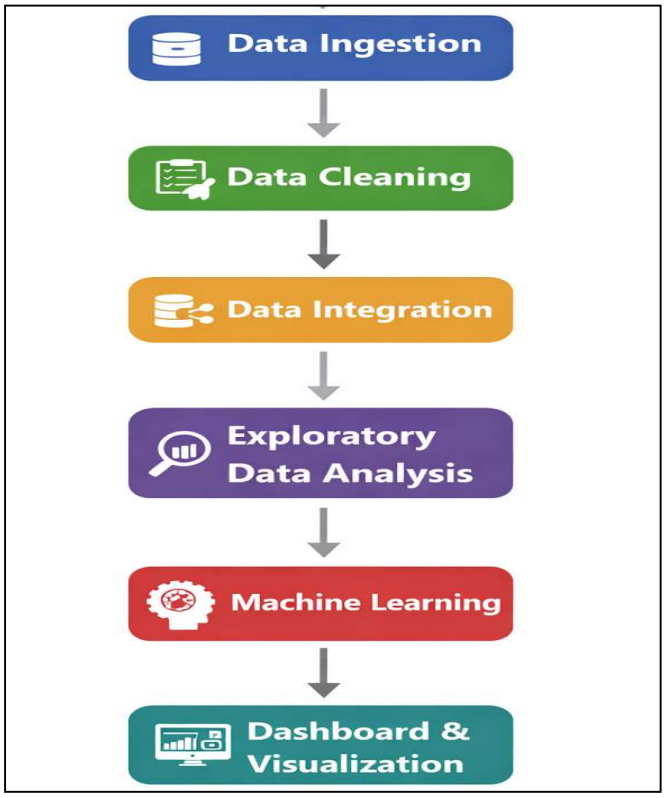

Fig. 5.1 Steps in method of implementation

5.1.1 Data Ingestion

Data ingestion is the initial and one of the most critical stages in the proposed system, as it involves collecting, organizing, and storing large volumes of raw data from multiple sources into a structured format. In this project, vendor-related data is gathered from multiple datasets and stored in a centralized SQLite database to enable efficient processing and analysis. Since the datasets are very large, containing millions of records, a chunk-based ingestion approach is used to load the data in smaller portions. This improves performance, prevents memory overflow, and ensures smooth handling of large-scale data.

Purchases Dataset:

The purchases dataset is one of the primary data sources used in this project, containing approximately 2.37 million records. It includes detailed information about purchase transactions such as inventory identifiers, store details, product descriptions, vendor information, and purchase order details. Additionally, it contains financial attributes such as purchase price, quantity, and total cost. This dataset plays a crucial role in understanding vendor supply patterns, procurement activities, and cost analysis.

Purchase Prices Dataset:

The purchase prices dataset consists of over 12,000 records and provides detailed information about product pricing offered by different vendors. It includes attributes such as product brand, description, size, volume, and purchase price along with vendor details. This dataset is essential for analyzing price variations among vendors and identifying cost-effective procurement strategies.

Vendor Invoice Dataset:

The vendor invoice dataset contains around 5,500 records and includes financial transaction details between vendors and the organization. It consists of information such as invoice dates, purchase order dates, payment dates, quantities, total transaction amounts, and freight charges. This dataset is useful for analyzing vendor payment cycles, additional costs, and overall financial interactions with vendors.

Beginning Inventory Dataset:

The beginning inventory dataset contains over 200,000 records and represents the stock levels at the start of a specific period. It includes details such as inventory identifiers, store locations, product descriptions, stock availability, pricing, and start dates. This dataset helps in understanding the initial inventory position and availability of products.

End Inventory Dataset:

The ending inventory dataset contains more than 220,000 records and represents the stock levels at the end of a given period. It includes similar attributes as the beginning inventory dataset. By comparing beginning and ending inventory datasets, it becomes possible to track inventory movement and calculate important metrics such as stock turnover.

Sales Dataset:

The sales dataset is the largest dataset used in this project, containing over 12.8 million records. It includes detailed information about sales transactions such as product details, sales quantity, sales revenue, sales price, and vendor information. This dataset is crucial for analyzing revenue generation, customer demand patterns, and the contribution of vendors to overall sales performance.

5.1.2 Data Preprocessing

Data preprocessing is a crucial step in the proposed system, as it ensures that the raw data collected during the ingestion phase is clean, consistent, and suitable for further analysis and modeling. Since the datasets used in this project are large and originate from multiple sources, they may contain inconsistencies such as missing values, duplicate records, incorrect formats, and noisy data. Therefore, preprocessing is performed to enhance data quality and reliability.

Data Cleaning

Data cleaning involves identifying and correcting errors present in the datasets. This includes handling missing values by either removing or imputing them, eliminating

duplicate records, and correcting inconsistent or invalid entries. Special attention is given to numerical fields such as purchase price, sales values, and quantities to ensure accuracy. Additionally, outliers are detected and treated carefully to avoid distortion in analysis results. This step ensures that the dataset is free from inconsistencies and is reliable for further processing.

Data Transformation

Data transformation is performed to convert the data into a suitable format for analysis. This includes converting date columns such as purchase date, sales date, and payment date into proper datetime formats, enabling time-series analysis. Numerical fields are standardized where necessary to maintain consistency across datasets. Textual fields such as vendor names, product descriptions, and store locations are also standardized to avoid duplication due to variations in naming conventions.

Feature Engineering

Feature engineering plays a significant role in enhancing the analytical capabilities of the system. New features are derived from existing data to provide deeper insights into vendor performance. Metrics such as Gross Profit are calculated by subtracting purchase cost from sales revenue, while Profit Margin is computed to measure profitability percentage. Stock Turnover is derived to understand how efficiently inventory is being utilized, and the Sales-to-Purchase Ratio is calculated to evaluate the balance between demand and supply. These engineered features help in improving both analysis and machine learning model performance.

5.1.3 Data Integration

Data integration is the process of combining multiple datasets into a unified and comprehensive dataset for analysis. In this project, datasets such as purchases, sales, inventory, purchase prices, and vendor invoices are integrated using structured SQL queries. This step ensures that all relevant information related to vendor performance is brought together into a single dataset.

The integrated dataset, referred to as the vendor sales summary, provides a holistic view of vendor activities, including procurement, sales performance, inventory movement, and financial metrics. By combining these datasets, relationships between different variables can be effectively analyzed. For instance, it becomes possible to study how purchase prices influence sales profitability or how inventory levels impact sales performance.

5.1.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to gain insights into the dataset and understand underlying patterns, trends, and relationships. This step is essential for identifying key characteristics of the data before applying machine learning models. Various statistical techniques and visualization methods are used during this phase.

The analysis includes generating summary statistics such as mean, median, and standard deviation to understand data distribution. Visualization techniques such as histograms, bar charts, and scatter plots are used to represent relationships between variables. Correlation analysis is conducted to identify dependencies between features such as sales, profit, and inventory levels.

Through EDA, important insights are derived, such as identifying top-performing vendors based on sales and profit, detecting low-performing vendors, and analyzing seasonal trends in sales. Outliers and anomalies are also identified during this stage, which helps in improving data quality and model performance.

5.1.5 Machine Learning

Machine learning techniques are applied in this project to analyze vendor performance and generate predictive insights. Advanced algorithms such as Random Forest and XGBoost are selected due to their ability to handle large datasets and capture complex relationships between variables. These algorithms are particularly effective for structured data and provide high accuracy.

The dataset is divided into training and testing sets to evaluate model performance. The models are trained using historical data, allowing them to learn patterns and relationships between input features and target variables. During the evaluation phase, performance metrics such as accuracy are used to measure the effectiveness of the models.

Among the implemented models, the best-performing model achieved an accuracy of approximately 99.59 percent, indicating a high level of prediction reliability. These machine learning models help in identifying key factors that influence vendor performance and enable predictive analysis, which can be used for future decision-making.

5.1.6 Dashboard and Visualization

The final stage of the proposed methodology involves presenting the analyzed data and insights through interactive dashboards. Visualization plays a crucial role in making complex data understandable and accessible to users. In this project, Power BI is used to develop dynamic and interactive dashboards.

The dashboards provide visual representations of key performance indicators such as top vendors, profit margins, sales trends, inventory turnover, and vendor contribution. Users can interact with the dashboard by applying filters, drilling down into specific data points, and exploring trends over time.

visualizations help stakeholders quickly understand business performance and make informed decisions. For example, managers can identify high-performing vendors, detect inefficiencies, and optimize procurement strategies. The use of dashboards enhances the overall usability of the system and ensures that insights are effectively communicated.

5.2 IMPLEMENTATION OF ALGORITHMS

The proposed vendor performance analytics system uses a combination of machine learning and data analysis techniques to evaluate vendor performance and generate

meaningful insights. The project mainly utilizes XGBoost for predictive analysis and K-Means clustering for grouping vendors based on performance patterns.

K-Means Clustering:

K-Means clustering plays an important role in the system by enabling effective vendor segmentation, where vendors are grouped into different categories based on their performance levels. This grouping helps in identifying performance trends by revealing patterns such as consistently high-performing vendors, average performers, and underperforming vendors. By analyzing these clusters, organizations can gain deeper insights into vendor behavior and performance variations. This ultimately supports better decision-making, allowing businesses to focus on reliable vendors, improve relationships with potential vendors, and take corrective actions for low-performing ones, thereby enhancing overall efficiency and profitability.

XGBoost:

XGBoost (Extreme Gradient Boosting) is a powerful machine learning algorithm based on gradient boosting techniques. It builds models sequentially, where each new model corrects the errors of the previous one.

In this project, XGBoost is used to predict vendor performance based on features such as purchase quantity, sales, profit margin, stock turnover, and sales-to-purchase ratio. It is highly efficient for large datasets and provides excellent performance in terms of accuracy and speed. The model achieved very high accuracy, making it suitable for real-world business decision-making.

5.3 OUTPUT SCREENS

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02

Fig. 5.2 summary statistics-1

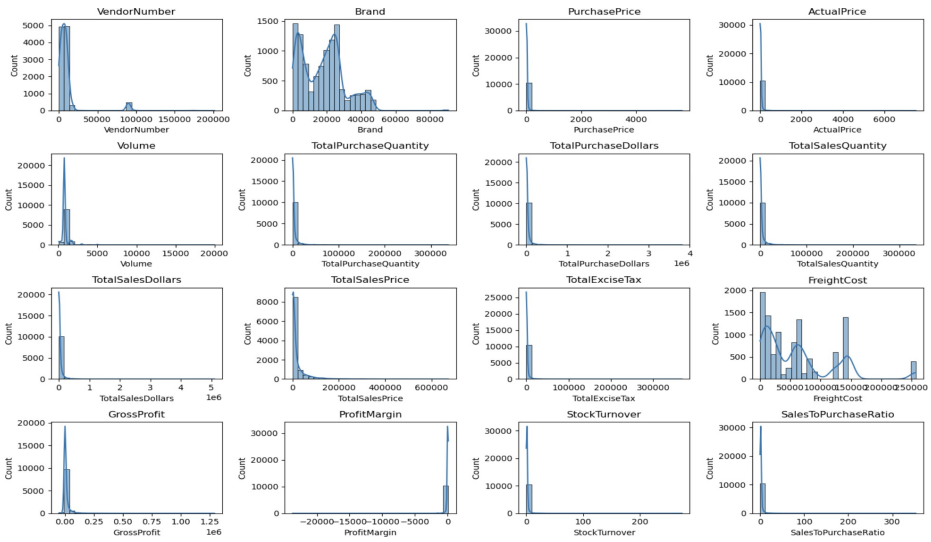

Fig. 5.3 summary statistics-2

Negative & Zero Values:

Gross Profit: Minimum of -52,002.78, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Has a minimum of -∞, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.

Total Sales Quantity & Sales Dollars: Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

Freight Cost: Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

Stock Turnover: Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than 1 indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

Correlation Insights:

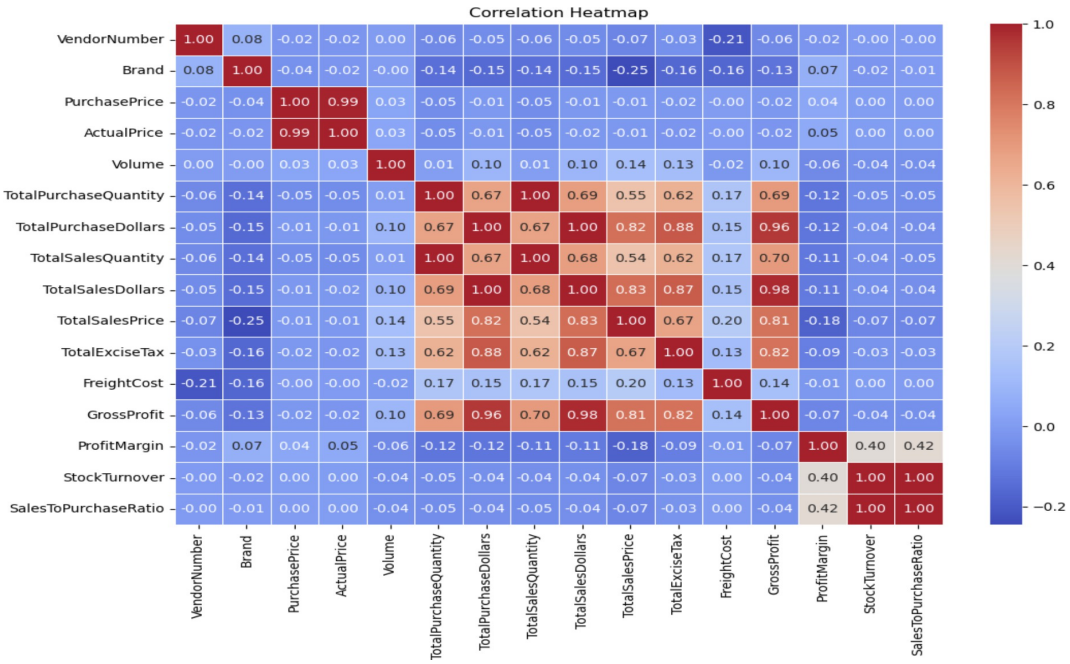

Fig. 5.4 Correlation Insights

Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong correlation (0.999), confirming efficient inventory turnover.

Profit Margin vs. Total Sales Price: Negative correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

Brands for Promotional or Pricing Adjustments:

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

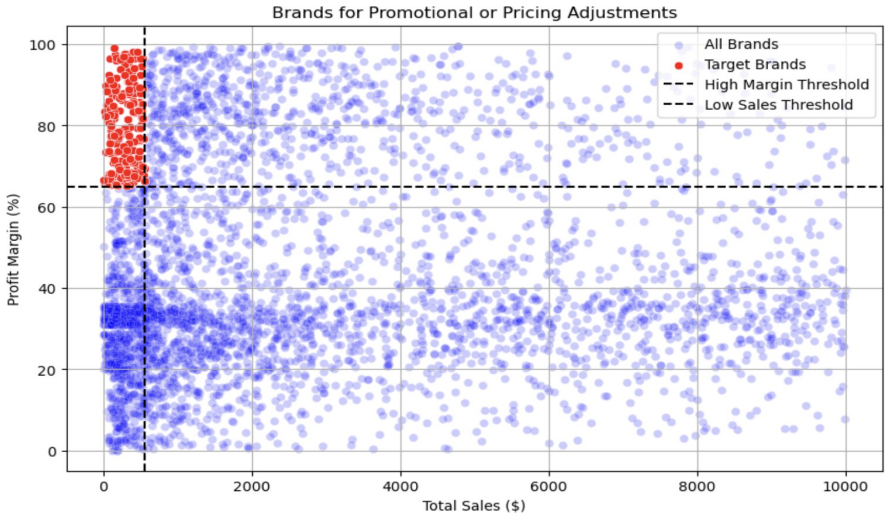

Fig. 5.5 Brands for Promotional or Pricing Adjustments

Top Vendors by Sales & Purchase Contribution:

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.

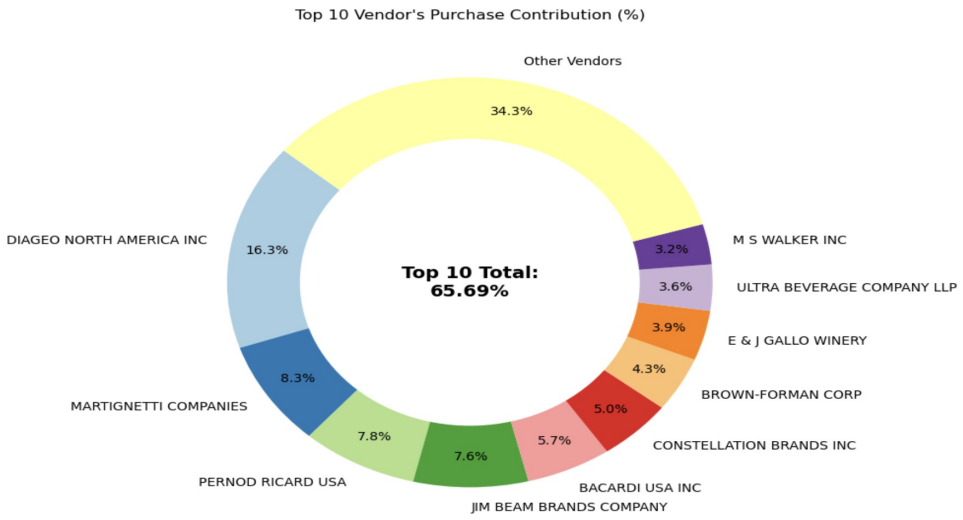

Fig. 5.6 Top Vendors by Sales & Purchase Contribution

Vendor Segmentation(Low/medium/high):

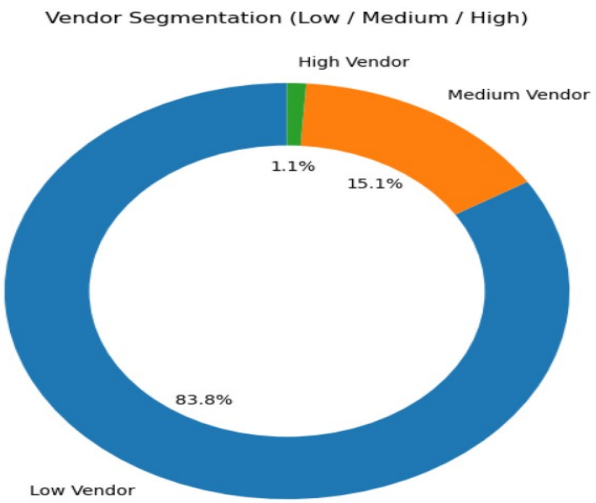

Fig. 5.7 Vendor Segmentation

The vendor segmentation chart classifies vendors into Low, Medium, and High performance categories using K-Means clustering. It shows that a majority of vendors fall under the low-performance category (83.8%), while a smaller portion are medium (15.1%) and very few are high-performing (1.1%).

Vendor Risk segementation:

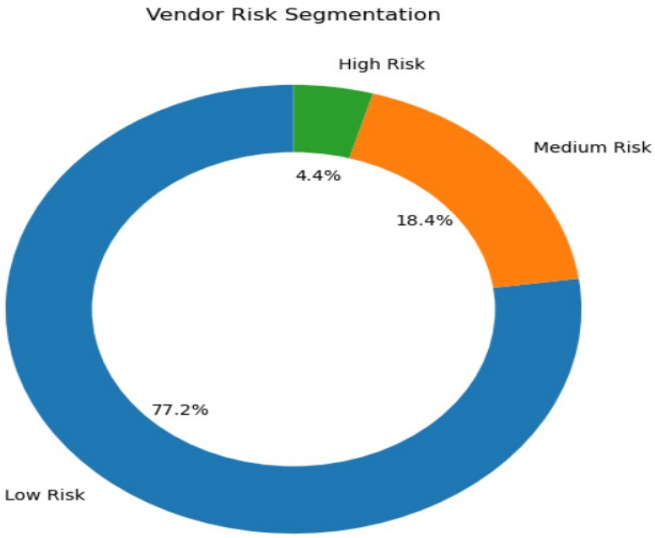

Fig. 5.8 Vendor Risk segementation

The vendor risk segmentation chart categorizes vendors into low, medium, and high-risk groups based on their performance and reliability. It indicates that most vendors are low risk (77.2%), while 18.4% fall under medium risk and only 4.4% are high-risk vendors. This helps businesses focus on minimizing risks and improving vendor reliability.

5.4 RESULT ANALYSIS

The results of the project provide a comprehensive analysis of vendor performance using large-scale datasets and machine learning techniques. The system processes data related to purchases, sales, inventory, and vendor transactions to generate meaningful insights. Using K-Means clustering and XGBoost, the system identifies vendor performance patterns, segments vendors, and highlights key metrics such as profit margin, stock turnover, and sales trends. The final results are visualized through interactive dashboards, enabling better decision-making.

5.4.1 Input

The system begins with data input in the form of multiple structured datasets, including

purchases, sales, inventory, purchase prices, and vendor invoices. These datasets are collected and stored in a SQLite database for efficient processing. Each dataset contains important attributes such as vendor details, product information, purchase quantities, sales values, and pricing data.

After data ingestion, preprocessing is performed to clean the data, handle missing values, and ensure consistency. Feature engineering is applied to generate key performance metrics such as Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio. Based on this processed data, new aggregated datasets (CSV files), such as vendor_sales_summary, are generated.

These resulting CSV files act as the final input for the visualization stage, where they are loaded into Power BI. This allows the system to transform processed data into interactive dashboards, enabling clear analysis of vendor performance, segmentation, and business insights.

5.4.2 Output

The output of the proposed system is presented in the form of an interactive and insightful dashboard developed using Power BI. After processing the input datasets through data cleaning, integration, and analytical techniques, the system generates summarized CSV files, which are then loaded into Power BI for visualization. The dashboard displays key business metrics such as Total Sales, Total Purchase, Gross Profit, Profit Margin, and Unsold Capital, providing a clear overview of organizational performance.

In addition to summary metrics, the dashboard includes detailed visualizations such as vendor segmentation (low, medium, high performance), vendor risk analysis, top vendors and brands by sales, purchase contribution percentages, and identification of low-performing brands using scatter plots. These visual insights enable users to easily interpret complex data patterns, identify trends, and make data-driven decisions. Overall, the system transforms raw data into meaningful business intelligence, supporting efficient vendor management and strategic planning.

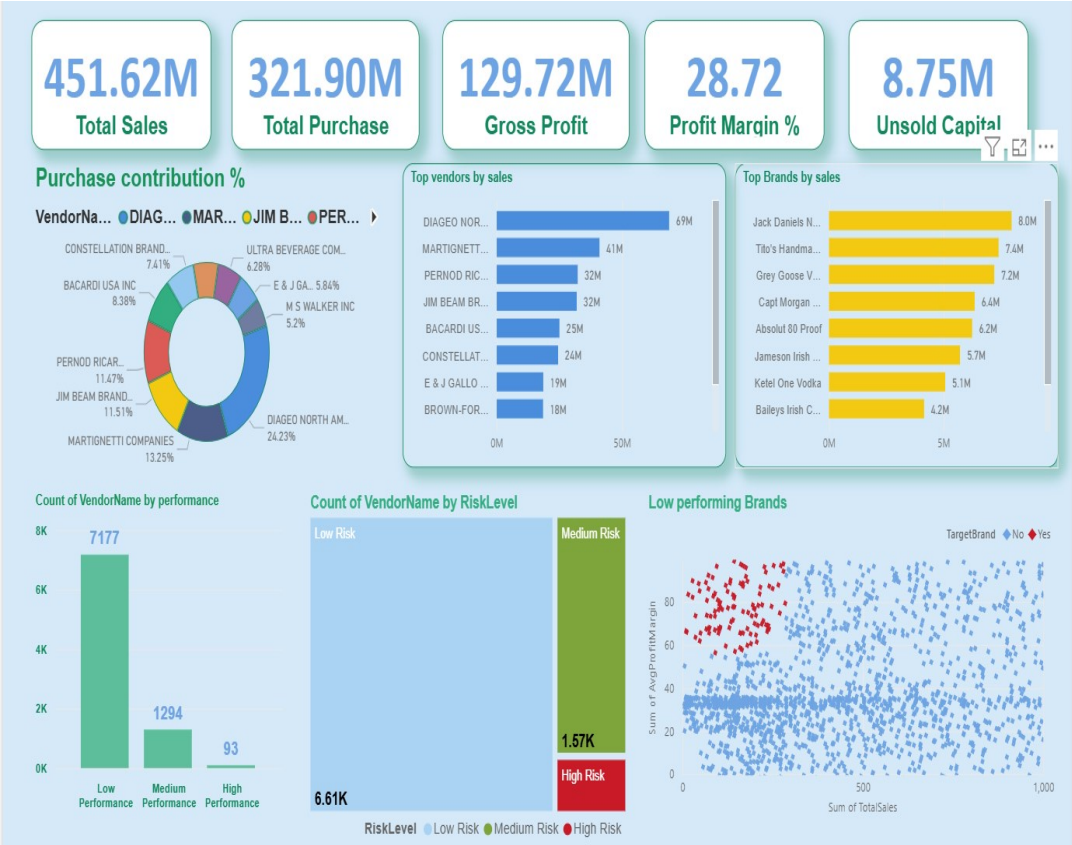

Fig. 5.9 Output Dashboard

6. TESTING & VALIDATION

6.1 FEATURES TO BE TESTED

Data Ingestion Features: The system's ability to ingest large datasets from multiple sources into the SQLite database is tested to ensure data integrity and completeness.

Data Preprocessing Features: Data preprocessing features such as cleaning, transformation, and feature engineering are validated to confirm that the data is consistent and analysis-ready.

Data Integration Features: Data integration processes are tested to ensure accurate merging of datasets and correct generation of the vendor_sales_summary table.

Analytical Features: Analytical features such as vendor performance calculation, clustering using K-Means, and identification of high and low-performing vendors are thoroughly tested.

Output and Visualization Features: The generation of output files in CSV format is verified to ensure correctness and compatibility with Power BI. The visualization layer is tested to confirm that dashboards accurately represent insights, support interactivity, and enable effective decision-making.

6.2 FEATURES NOT TO BE TESTED

Real-Time Processing: Certain functionalities such as real-time data processing and streaming are not tested, as the system operates on batch-processed datasets.

External Integrations: External system integrations such as ERP systems or third-party APIs are not tested, as the project focuses solely on internal data analysis.

Security Features: Security-related features such as user authentication and authorization are not included since the system is designed as an analytical tool rather than a user-driven application.

Cloud and Deployment Features: Cloud deployment and distributed computing environments are also not considered in testing.

Advanced Models: Advanced deep learning models are not part of this project, so their testing is excluded.

6.3 TESTING TOOLS AND ENVIRONMENT

Programming Environment: The testing of the Vendor Performance Analytics System is conducted using Python (Version 3.x) as the primary programming language.

Libraries and Tools: Libraries such as Pandas and NumPy are used for data manipulation, while Matplotlib and Seaborn are used for validating visual outputs. Scikit-learn is used for implementing and testing K-Means clustering and analytical techniques.

Database Testing: SQLite is used as the database, and SQL queries are tested to ensure correct data integration and processing.

Development Environment: The system is developed and tested using tools such as Jupyter Notebook and Visual Studio Code, which support debugging and validation.

Visualization Testing: Power BI is used to test the final visualization layer, ensuring that dashboards correctly display insights and support user interaction.

System Compatibility: The system is tested on standard operating systems such as Windows to ensure compatibility and smooth execution.

6.4 DESIGN OF TEST CASES AND SCENARIOS

Data Ingestion Testing: The design of test cases and scenarios in the Vendor Performance Analytics System focuses on validating each stage of the workflow, from data ingestion to final visualization. Test cases are created to ensure that all datasets such as purchases, sales, inventory, vendor invoices, and purchase prices are correctly loaded into the system without data loss or corruption.

Data Preprocessing Testing: During data preprocessing, test scenarios verify the handling of missing values, removal of duplicates, correction of inconsistent data formats, and proper feature engineering.

Data Integration Testing: In the data integration stage, test cases are designed to validate the correctness of SQL joins and aggregations used to generate the vendor_sales_summary dataset.

Analysis and Modeling Testing: For the analysis phase, test cases ensure that K-Means clustering correctly segments vendors and that calculated metrics such as Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio are accurate.

Output and Visualization Testing: In the output and visualization stage, test scenarios validate the correctness of generated CSV files and ensure that Power BI dashboards display accurate insights such as top vendors, low-performing vendors, and sales trends. Performance testing is also conducted to verify the system's efficiency when handling large datasets.

7. CONCLUSION & FUTURE ENHANCEMENTS

7.1 CONCLUSION

The proposed vendor performance analytics system adopts a comprehensive and data-driven approach by integrating data analytics, machine learning, and visualization techniques to evaluate and optimize vendor performance. The system utilizes multiple datasets, including purchases, sales, inventory, and vendor-related information, which are processed and transformed into meaningful insights through data preprocessing, feature engineering, and data integration. Key performance indicators such as Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio are derived to enable effective vendor evaluation.

Clustering techniques such as K-Means are employed to segment vendors based on their performance and risk levels, enabling better identification of low, medium, and high-performing vendors. Additionally, machine learning models are implemented to analyze patterns and support predictive insights, achieving a high accuracy of 99.59%. The integration of Power BI dashboards further enhances the system by providing interactive and intuitive visualizations, including vendor segmentation, risk analysis, top vendors, and sales trends. Overall, the system successfully transforms raw business data into actionable insights, supporting efficient decision-making, cost optimization, and improved vendor management

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7.2 FUTURE ENHANCEMENTS

Advanced Machine Learning Models: The proposed system can be further enhanced by integrating advanced machine learning and deep learning models to improve prediction accuracy and identify more complex patterns in vendor performance and business trends.

Real-Time Data Processing: Future improvements may include implementing real-time data processing to enable continuous monitoring of vendor performance and instant generation of insights for faster decision-making.

Automated Decision-Making: The system can be extended to include automated decision-making capabilities that provide intelligent recommendations for vendor

selection, inventory management, and cost optimization.

AI-Based Chatbot Integration: An AI-powered chatbot can be integrated into the system to allow users to interact using natural language queries, making the system more user-friendly and interactive.

Sales Forecasting: Incorporating forecasting models can help predict future sales trends, enabling better inventory planning and demand management.

Web and Mobile Deployment: Deploying the system as a web or mobile application will enhance accessibility, allowing users to access insights anytime and anywhere, making it a complete business intelligence solution.

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